

NATO's approach to space

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Space is a dynamic and rapidly evolving environment, which is essential to the Alliance's deterrence and defence. NATO recognises space as an operational domain, alongside air, land, maritime and cyberspace. Allies have agreed an Overarching Space Policy to guide NATO's approach to space and ensure the right space-based support to the Alliance's operations, missions and other activities in areas such as communications, navigation and intelligence.

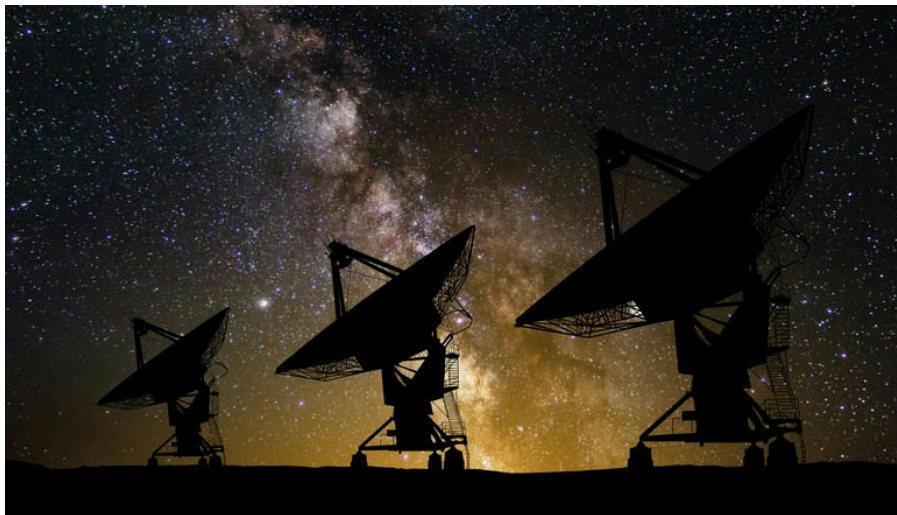


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- Space is increasingly important for both civilian and military activities. Space capabilities bring benefits in multiple areas – from weather monitoring, environment and agriculture, to transport, science, communications and banking. They are also critical for NATO operations, missions and other activities, including collective defence, crisis response, intelligence-gathering and counter-terrorism.
- At the same time, space is becoming more crowded and competitive, and

and China, have developed and tested a wide range of counter-space technologies.

- In this context, NATO serves as a key forum for Allies to share information and coordinate activities on various space-related issues.
- In 2019, Allies adopted an Overarching [Space Policy](#) and declared space an operational domain.
- In 2020, Defence Ministers decided to establish a NATO Space Operations Centre at Allied Air Command in Ramstein, Germany.
- At the 2021 Brussels Summit, NATO recognised that attacks to, from or within space present a clear challenge to the security of the Alliance and could lead to the invocation of Article 5 of the North Atlantic Treaty.
- In July 2024, Defence Ministers from 17 NATO Allies signed the Memorandum of Understanding for the Alliance Persistent Surveillance from Space (APSS) programme to enhance space-based surveillance and intelligence for the Alliance.
- In February 2025, Defence Ministers endorsed the NATO Commercial Space Strategy, aiming to strengthen the Alliance's relationship with the commercial space sector and improve NATO's ability to leverage the growing array of commercial space solutions.
- At the 2026 NATO Summit Defence Industry Forum in Ankara, eight NATO Allies launched the Hybrid Alliance Layered Operations in Space (HALO) initiative, which aims to improve resilience and military advantage in space by connecting and integrating sovereign, nationally owned and controlled military satellites into a networked mega-constellation.
- NATO's approach to space will remain fully in line with international law.

Why is space important for NATO and Allies?

Space is essential to the Alliance's deterrence and defence, underpinning its ability to navigate and track forces, to maintain robust communications, to

detect missile launches and to ensure effective command and control. The use

respond to crises with greater speed, effectiveness and precision. Space data, products and services are a critical enabler of NATO operations and directly support activities in other operational domains (air, land, maritime and cyberspace).

From a security and defence perspective, space is critical for the Alliance, including in the following areas:

- **positioning, navigation and timing**, which enables precision strikes, tracking of forces, and search and rescue missions;
- **early warning**, which helps to ensure force protection and provides vital information on missile launches;
- **environmental monitoring**, which enables meteorological forecasting and mission planning;
- **secure satellite communications**, which are essential for missions to enable consultation, command and control;
- **intelligence, surveillance and reconnaissance**, which are crucial for situational awareness, planning and decision-making.

NATO Allies increasingly rely on space for various national security tasks, as well as military operations around the globe. More than half of active satellites orbiting the Earth belong to NATO members or companies based on their territory.

How is the security environment in space changing?

Today, access to and use of space is no longer limited to a few countries that are technically capable of launching and operating a spacecraft. Space technology and services have become cheaper to produce and operate, more capable and more readily accessible to a wider range of countries and private actors. In security and defence terms, space is increasingly contested, congested and competitive and now requires the Alliance to be able to operate in a disrupted,

risks, vulnerabilities and potential threats. While space can be used for peaceful purposes, it can also be used for aggression. In addition, satellites can be hacked, jammed or weaponised, and anti-satellite weapons could cripple communications and affect the Alliance's ability to operate.

Some countries, including Russia and China, have developed and tested a wide range of counter-space technologies that could restrict Allies' access and freedom to operate in space. These developments in space capabilities and activities are cause for concern. They represent increasing risks to Allied space systems, including the potential targeting of both civilian and military infrastructure, which could cause harm to both security and commercial interests.

Furthermore, the potential stationing of weapons in space poses an even greater threat. NATO Allies oppose any placement of nuclear weapons in orbit around Earth, which would violate Article IV of the Outer Space Treaty, and would gravely threaten global security.

NATO's approach to space will remain fully in line with international law. The Alliance has no intention to put weapons in space. NATO Allies call on other states to support international efforts to promote responsible space behaviour.

NATO's space-related activities

NATO is an important forum for Allies to share information, increase interoperability and coordinate actions. NATO is not aiming to develop space capabilities of its own and relies on Allies' space assets.

In 2019, Allies adopted an [Overarching Space Policy](#) and declared space an operational domain. To implement this Policy, the Alliance is pursuing activities across several lines of effort:

- Space support in operations, missions and other activities;
- Space domain awareness (i.e. enhancing Allies' common understanding of the space environment, including threats and risks);
- Deterrence, defence and resilience;

- Training and exercises;
- Strategic communications and responsible space behaviours;
- Science, technology and innovation;
- Cooperation with industry and the commercial sector;
- Cooperation with partner countries and relevant international organisations, including the European Union and the United Nations.

Below are some specific examples of initiatives that NATO has developed under these lines of effort.

The NATO Space Operations Centre

The [NATO Space Operations Centre \(NSpOC\)](#) was established in 2024 at Allied Air Command in Ramstein, Germany, as part of the Combined Force Space Component Command (CFSpCC). It serves as a focal point to support NATO's operations, missions and other activities; share information; and help coordinate Allies' efforts in the space domain.

The NSpOC reaches out to national space entities to ensure that NATO commanders have access to required space data and services. Streamlining requests for space products through a single entity increases NATO's responsiveness and supports timely decision-making. The NSpOC closely follows the developments in the space security environment.

NATO's engagement with the commercial space sector

Recognising that technological advancement in space is no longer limited to state actors, NATO has set out to strengthen its relationship with the commercial space sector and improve its ability to leverage the growing array of commercial space data, products and services.

To that end, Allied Defence Ministers endorsed the first [NATO Commercial Space Strategy](#) in February 2025. The strategy aims to better leverage the use of commercial solutions; ensure that commercial space services are readily

consultation with industry partners, the strategy was publicly released at the 2025 NATO Summit Defence Industry Forum during the 2025 NATO Summit in The Hague.

Furthermore, under the NATO Industrial Advisory Group (NIAG), the Alliance has created a new network called SPACENET to foster a closer relationship with space industry. SPACENET allows industry actors to provide NATO and Allies with insights into novel technological developments and market offers, and advise the Alliance on challenges such as interoperability and standardization issues. SPACENET is open to all space companies from across the Alliance. For questions and further guidance please reach out [via the NIAG](#).

Delivering space capabilities for the Alliance

NATO is stepping up its efforts to support Allied delivery of cutting-edge space technologies, particularly in the following areas:

NATO SATCOM Services 6th Generation (NSS6G)

To allow NATO forces to communicate more securely and quickly, NATO is investing over EUR 1 billion in procuring [satellite communications](#) services for the period of 2020-2034 through the NATO SATCOM Services 6th Generation (NSS6G) project. This major investment in satellite communications is being provided by NATO member countries. NSS6G enables more resilient and flexible communications with ships at sea, air assets and troops across the globe.

Strategic Space Situational Awareness System (3SAS)

NATO is developing a Strategic Space Situational Awareness System (3SAS) to allow the Alliance to better understand the space environment and space events, and their effects across all domains. The 3SAS initiative was [launched in June 2021](#), and is supported by funding from Luxembourg.

Alliance Persistent Surveillance from Space (APSS)

While 3SAS focuses on looking up at space, it's equally important for the Alliance

happening on the ground. To that end, NATO is currently implementing APSS – the largest multinational investment in space-based capabilities in NATO's history. APSS will enhance cooperation on space-based surveillance, establishing a large-scale, virtual constellation of national and commercial surveillance satellites, called "Aquila".

The APSS initiative was [launched in February 2023](#) and has now [transitioned into the implementation phase](#). Nineteen Allies are contributing the equivalent of more than USD 1 billion to leverage commercial and national space assets, and to expand advanced exploitation capacities. The participating NATO Allies are: Belgium, Bulgaria, Canada, Denmark, Finland, France, Germany, Greece, Hungary, Italy, Luxembourg, the Netherlands, Norway, Poland, Romania, Spain, Sweden, Türkiye and the United States.

NORTHLINK

Allies are currently exploring the development of a secure, resilient and reliable Arctic satellite communications capability. NORTHLINK is intended to leverage commercial and government-owned capabilities, providing more resilient communications in this region. The NORTHLINK project was [launched in October 2024](#). It currently includes 13 NATO Allies: Canada, Denmark, Finland, France, Germany, Hungary, Iceland, Italy, Luxembourg, the Netherlands, Norway, Sweden and the United States.

STARLIFT

Allies are investigating ways to strengthen NATO's access to and use of space in support of the Alliance's deterrence and defence. STARLIFT aims to develop a more resilient, responsive and cost-effective network of launch capabilities that will help Allies launch assets at short notice from space ports across the Alliance. The STARLIFT project was [launched in October 2024](#) and currently includes 15 NATO Allies: Belgium, Canada, Finland, France, Germany, Hungary, Italy, Luxembourg, the Netherlands, Norway, Spain, Sweden, Türkiye, the United Kingdom and the United States.

Hybrid Alliance Layered Operations in Space (HALO)

Allies are working on ways to improve connectivity, resilience and military advantage in space. The multinational HALO initiative aims to develop a networked mega constellation that connects sovereign, nationally owned and controlled military satellites of individual Allies. Its goal is to enable high-speed communications, intelligence and missile tracking, and overcome the cost, time and coverage limitations of single-nation satellite fleets. The initiative was [launched at the 2026 NATO Summit Defence Industry Forum in Ankara](#) and currently includes eight NATO Allies: Canada, Denmark, Finland, Germany, the Netherlands, Norway, Sweden and Türkiye.

Exercises and innovation

The Alliance is also integrating space in training and exercises, operational planning, capability development, as well as in its innovation efforts. [Emerging and disruptive technologies](#) are transforming the space domain and NATO will take advantage of these developments to maintain its technological edge. In this regard, NATO's Science and Technology Organization network helps to leverage the scientific and technological capacity among Allies and partners.

Evolution

NATO Allies have recognised the importance of space for security and defence since the Alliance's creation in 1949.

Throughout NATO's history, space capabilities have largely been a national responsibility, but the Alliance did [launch eight communications satellites](#) between 1970 and 1993. NATO operated these satellites until 2005, when the [NATO SATCOM Post-2000 \(NSP2K\)](#) programme replaced the NATO-owned satellites with a system where the Alliance had direct access to national communications satellites from France, Italy and the United Kingdom.

At the [2018 Brussels Summit](#), NATO Leaders agreed to develop an Overarching Space Policy to guide the Alliance's activities in this highly dynamic and rapidly evolving area.

In June 2019, NATO Defence Ministers adopted [NATO's Overarching Space Policy](#).

At the [2019 Leaders' Meeting in London](#), Allies declared space a fifth operational domain, alongside air, land, maritime and cyberspace. In their declaration, NATO Leaders stated: "We have declared space an operational domain for NATO, recognising its importance in keeping us safe and tackling security challenges, while upholding international law."

As of January 2020, the NSP2K programme was succeeded by a [new satellite services project](#) called NATO SATCOM Services 6th Generation, or NSS6G.

[In October 2020](#), Defence Ministers decided to establish a NATO Space Operations Centre at Allied Air Command in Ramstein, Germany. It is part of the Combined Force Space Component Command (CFSpCC).

At the [2021 Brussels Summit](#), NATO Leaders stated that "attacks to, from, or within space present a clear challenge to the security of the Alliance, the impact of which could threaten national and Euro-Atlantic prosperity, security, and stability, and could be as harmful to modern societies as a conventional attack. Such attacks could lead to the invocation of Article 5. A decision as to when such attacks would lead to the invocation of Article 5 would be taken by the North Atlantic Council on a case-by-case basis."

In November 2021, NATO Allies condemned Russia's reckless and irresponsible [anti-satellite missile test](#) of 15 November 2021. This test caused an orbital debris field that significantly increased risk to human life and to the space-based assets of numerous countries and entities. This dangerous behaviour directly contradicted Russia's claims to oppose the "weaponisation" of space, and undermined the rules-based international order. NATO Allies remain committed to protecting and preserving the peaceful access to and exploration of space for all humanity.

At the 2022 Madrid Summit, NATO Leaders adopted the [2022 Strategic Concept](#), the guiding document that defines the security challenges facing the Alliance and outlines the political and military tasks that NATO will carry out to address them. In addition to describing the increasing threats in space, the Strategic Concept also contains a commitment from Allies to "enhance our ability to operate effectively in space and cyberspace to prevent, detect, counter and respond to the full spectrum of threats, using all available tools... We will also boost the resilience of the space and cyber capabilities upon which we depend

In February 2023, a group of NATO Allies together with former Invitees Finland and Sweden (now NATO member countries) [agreed to develop](#) the Alliance Persistent Surveillance from Space (APSS) programme. Luxembourg provided an early contribution of EUR 16.5 million to lay the groundwork for this transformative initiative.

At the 2023 Vilnius Summit, NATO Leaders welcomed the progress made on integrating space into the Alliance's work through various initiatives, including the establishment of the [NATO-accredited Space Centre of Excellence](#) in Toulouse, France.

At the 2024 Washington Summit, NATO Leaders agreed to further accelerate the integration of space into NATO's planning, exercises and multi-domain operations, in particular by strengthening the capacity of NATO's Space Operations Centre located at Allied Air Command in Ramstein.

On the sidelines of the 2024 Washington Summit, [17 Allies signed a Memorandum of Understanding](#) transitioning the APSS project into its implementation phase.

In February 2025, Defence Ministers agreed the [NATO Commercial Space Strategy](#), which lays the groundwork for a strengthened and continuous relationship between NATO and Allies as well as the commercial space sector. It was developed in consultation with industry partners.

At the 2026 NATO Summit Defence Industry Forum in Ankara, [eight Allies launched the Hybrid Alliance Layered Operations in Space \(HALO\) initiative](#), which aims to improve connectivity and resilience by integrating sovereign, nationally owned and controlled military satellites into a networked mega constellation.

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